A Probability Metric-Based Autoencoder for Hyperspectral Unmixing

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***Abstract*— In recent years, hyperspectral unmixing (HU) has been a crucial preprocess in hyperspectral imagery analysis. Current learning-based methods of HU are built on the autoencoder framework. Most of them hold a pointwise loss function only comparing the individual difference without the overall difference. As a remedy, we propose a probability metric-based autoencoder (PMAE) for HU. Specifically, we combine a distribution distance called maximum mean discrepancy (MMD) with underlying loss spectral angle distance (SAD) as the final loss function for training the unmixing autoencoder. The MMD can be seen as a regularization that is designed to match all orders of statistics between the input spectrums and the reconstructed spectrums. The SAD is to compare the individual difference, while the MMD compares the overall difference. Combing these two points, we can obtain more prior information to guide the autoencoder training toward boosting the unmixing performance. In order to validate this viewpoint, we conduct model analysis and a series of ablation studies, showing that adding the MMD into the autoencoder unmixing model, indeed, boosts the unmixing performance a lot compared to the single SAD objective. Experimental results on five real hyperspectral scenes demonstrate the effectiveness and competitiveness of the proposed PMAE compared with several classical and learning-based approaches.**

***Index Terms*— Autoencoder network, hyperspectral unmixing (HU), maximum mean discrepancy (MMD), spectral angle distance (SAD).**

I. Introduction

PAST decades have witnessed the rapid development of hyperspectral imaging in the field of remote sensing for its ability to access rich spectral information. Benefiting from the wealthy spectral information, hyperspectral imageries (HSIs) have many practical applications, such as land cover classification, data fusion, and anomaly detection [1]–[7]. However, due to the relatively low spatial resolution of sensors and complex interactions between substances, each pixel of the HSIs is often a mixture of several materials, which inevitably affects the performance of upper level tasks. Hyperspectral unmixing (HU) has emerged as a strategy to tackle this issue intended to decompose the mixed pixel into a linear combination of several materials, i.e., endmember and their corresponding weights called abundance vector. Based on linear mixing models (LMMs) and nonlinear mixing models [8], various unmixing algorithms can be derived. In practice, most unmixing methods employ the LMMs due to being simple yet effective.

Traditional linear unmixing approaches can be divided into two categories, namely, pure-pixel-based and nonpure-pixel-based. The pure-pixel-based methods take the hypothesis that there exists at least one pure pixel in the scene, the unmixing process amounts to search the vertices of the simplex constructed by the hyperspectral dataset, and these vertices are considered as endmembers. Some classical methods based on pure pixel are the pixel purity index (PPI) [9], N-FINDR [10], and vertex component analysis (VCA) [11]. However, this assumption will be no longer available when the spatial resolution of the HSIs is very low. Meanwhile, such methods require additional methods, such as FCLS [12] for abundance estimation. What is more, the performance of the estimated abundance depends on the quality of the extracted endmembers.

As a remedy, nonpure-pixel-based methods emerged, including volume minimization approaches [13]–[16] and nonnegative matrix factorization (NMF)-based methods [17]–[20]. Among them, NMF is a popular research trend for traditional practitioners due to its flexibility of unsupervised unmixing. Qian *et al*. [17] presented the so-called $L_{1/2}$-NMF by imposing $L_{1/2}$-norm on the abundance matrix. Lu *et al*. [18] first introduced manifold regularization into sparsity-constrained NMF for the HU task. Li *et al*. [19] developed a robust collaborative NMF, which could estimate the number of endmembers via an additional collaborative regularization. Hong *et al*. [20] considered integrating both scaling factors and spectral variabilities into the NMF model. However, NMF-based methods may suffer from that the nonconvexity of the objective function leads to falling into local minimum solutions and the high dependence on the initialization.

As an alternative to the aforementioned methods, the researchers have proposed learning-based methods for the HU task, namely, in the form of an autoencoder [21]. Concretely, the encoder is to compress the input spectra into a potential representation, i.e., an abundance vector. At the same time, the decoder reconstructs the input spectra from the representation via a linear mapping, which is considered as endmembers. Then, one can obtain abundances and endmembers by feeding all training samples into the autoencoder when the training is finished. This class of methods focuses on the study of network frameworks and objective functions. Palsson *et al*. [22] proposed a multilayer autoencoder for the HU task and compared the unmixing performance differences

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among losses: the spectral angle distance (SAD), the mean square error (mse), and the spectral information divergence (SID). One can find that SAD can obtain superior results to the other two losses due to its scale invariance. Under a two-staged autoencoder network, Ozkan *et al*. [23] considered combining SAD with other penalty terms, such as mse and L_1 sparsity, as the objective function to improve the unmixing performance. Su *et al*. [24] adopted stacked autoencoders to generate a reliable initialization for the subsequent variational autoencoder, which is to perform NMF. Based on the observation that different abundance maps are close to orthogonal, Dou *et al*. [25] added an orthogonal sparse prior into a four-layer autoencoder network with the underlying hyper-Laplacian loss. Motivated by the convolutional neural network (CNN), Elkholy *et al*. [26] presented a 1-D deep convolutional autoencoder for the HU task, and in order to extract spatial–spectral information, Palsson *et al*. [27] constructed a 2-D deep convolutional autoencoder to conduct unmixing; they both employed the SAD as the objective function. Such methods have made great progress in the unmixing area, but there is still room for improvement.

It is worth noting that the aforementioned learning-based methods always possess the pointwise loss functions. Specifically, these loss functions have the form $(1/n)\sum_{i=1}^{n}\mathcal{L}(\boldsymbol{x}_i, \boldsymbol{y}_i)$ between the inputs $\{\boldsymbol{x}_i\}_{i=1}^{n}$ and the reconstructions $\{\boldsymbol{y}_i\}_{i=1}^{n}$; they only take the difference of a single sample pair $(\boldsymbol{x}_i, \boldsymbol{y}_i)$ into account for training. In this manner, we just capture the individual information but not overall information that contributes to improving the unmixing performance. Therefore, our work aims at designing a loss that can compare the overall difference. An intuitive idea is to consider comparing the high-order statistical differences between $\{\boldsymbol{x}_i\}_{i=1}^{n}$ and $\{\boldsymbol{y}_i\}_{i=1}^{n}$. From the mathematical perspective, we regard the input and the reconstruction as two distributions and then compare their distribution similarity. The work relevant to ours was proposed by Min *et al*. [28] who took the Wasserstein distance (WD) as the distribution similarity metric. However, it requires an additional network to implicitly compute the WD leading to the need for carefully designed training dynamics.

Conversely, we adopt the maximum mean discrepancy (MMD) as the distribution distance, which is interpreted as matching all orders of statistics between $\{\boldsymbol{x}_i\}_{i=1}^{n}$ and $\{\boldsymbol{y}_i\}_{i=1}^{n}$. Compared to WD, the MMD has the following two advantages that it would not suffer from the sample complexity when the data dimensionality is high, and it can be computed explicitly. Furthermore, the MMD originates from integral probability metrics (IPMs) [29], which has been applied to many areas, such as image generation and unsupervised domain adaptation [30]–[33]. These applications essentially are belonging to the generative model. Coincidentally, the autoencoder also can fall into this scope, so it is also natural to apply the MMD to train unmixing autoencoder. This motivation is very straightforward and simple, and our contributions can be summarized as follows.

1) In this article, we propose a new strategy to train the autoencoder unmixing model by integrating the MMD regularization with the underlying SAD as the total loss. The MMD regularization is designed to compare the distribution difference between the input spectrums and the reconstructed spectrums by matching their statistics; thereby, the unmixing model can capture the geometric nature to improve the performance.
2) We give a theoretical comparison of sample complexity between WD and MMD and show that adopting the MMD as the distribution distance is a more effective way when the data dimensionality is high.
3) We conduct parameter analysis to investigate the impacts of batch size, abundance sum-to-one (ASC) constraints, and MMD regularization on unmixing performance. The ablation studies show that adding the MMD regularization into the underlying SAD objective, indeed, can boost the unmixing performance a lot compared to the single SAD loss. The quantitative and qualitative results on five hyperspectral scenes show that the probability metric-based autoencoder PMAE) is effective and competitive.

The remainder of this article is organized as follows. Section II briefly reviews some basic background knowledge. Section III describes our proposed PMAE method in detail consisting of the network architecture and the loss function. Section IV provides parameter analysis, and quantitative and qualitative results to illustrate the effectiveness and competitiveness of the PMAE. Section V draws the conclusion.

II. Preliminary Knowledge

In this section, we will first review the LMM, then briefly elaborate on the IPM, and, finally, introduce two basic operations in the neural network.

A. Linear Mixing Model

Given an observed HSI $\boldsymbol{Y} \in \mathbb{R}^{B\times M\times N}$, it can be reshaped into $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p, \ldots, \boldsymbol{x}_P] \in \mathbb{R}^{B\times P}$, where B and $P = M \times N$ correspond to the number of bands and the number of pixels, respectively. The LMM assumes that the observed spectra $\boldsymbol{x}_p \in \boldsymbol{X}$ is a linear combination of several constituent endmembers, and their corresponding weights account for the fractional abundances. That is,

$$\boldsymbol{x}_p = \sum_{r=1}^{R} a_{r,p}\boldsymbol{e_r} + \boldsymbol{\varepsilon}_p = \boldsymbol{E}\boldsymbol{a}_p + \boldsymbol{\varepsilon}_p \tag{1}$$

where $\boldsymbol{e_r}$ is the rth endmember, $\boldsymbol{E} \in \mathbb{R}^{B\times R}$ represents the endmember matrix with R constituent materials, $\boldsymbol{a}_p = [a_{1,p}, \ldots, a_{R,p}]^T \in \mathbb{R}^{R\times 1}$ is the abundance vector, and $\boldsymbol{\varepsilon}_p \in \mathbb{R}^{B\times 1}$ stands for the noise vector. Then, under the matrix form, (1) can be reformulated as

$$\boldsymbol{X} = \boldsymbol{E}\boldsymbol{A} + \boldsymbol{\mathcal{E}} \tag{2}$$

where $\boldsymbol{\mathcal{E}} = [\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_p, \ldots, \boldsymbol{\varepsilon}_P] \in \mathbb{R}^{B\times P}$ is the noise matrix. To meet the physical meaning of unmixing, two constrains should be added to (2), i.e., abundance nonnegativity (ANC) and ASC. Formally, they are formulated as

$$\text{ANC: } \boldsymbol{A} \succeq 0 \tag{3a}$$

$$\text{ASC: } \mathbf{1}_R^T\boldsymbol{A} = \mathbf{1}_P^T \tag{3b}$$

where $\mathbf{1}$ is an all-ones vector, and $\boldsymbol{A} \succeq 0$ means that each element of matrix $\boldsymbol{A}$ is no smaller than zero.

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B. Integral Probability Metrics

IPMs [29] are a popular family of distance measures on probability used to measure the similarity between two probability distributions. Consider a compact space $\mathcal{X} \in \mathbb{R}^d$, and let $\mathcal{F}$ be a set of measurable and bounded real-valued function $\mathcal{X}$. Denote the $\mathcal{P}(\mathcal{X})$ as the set of measurable probability distribution on $\mathcal{X}$. Given two probability distributions $\mathbb{P}, \mathbb{Q} \in \mathcal{P}(\mathcal{X})$, the IPM induced by the function space $\mathcal{F}$ is defined as follows:

$$d_{\mathcal{F}}(\mathbb{P}, \mathbb{Q}) = \sup_{f \in \mathcal{F}} \left| \mathop{\mathbb{E}}_{\boldsymbol{x} \sim \mathbb{P}} f(\boldsymbol{x}) - \mathop{\mathbb{E}}_{\boldsymbol{y} \sim \mathbb{Q}} f(\boldsymbol{y}) \right|. \tag{4}$$

By choosing the suitable $\mathcal{F}$, various distances between probability measures can be derived. Choosing $\mathcal{F} = \{f : \|f\|_{\text{Lip}} \leq 1\}$ in (4) leads to the WD with the following expression:

$$\mathrm{W}(\mathbb{P}, \mathbb{Q}) = \inf_{\gamma \in \Pi(\mathbb{P}, \mathbb{Q})} \mathbb{E}_{(\boldsymbol{x}, \boldsymbol{y}) \sim \gamma} [\|\boldsymbol{x} - \boldsymbol{y}\|] \tag{5}$$

where $\|\cdot\|_{\text{Lip}}$ stands for the Lipschitz norm, and $\Pi(\mathbb{P}, \mathbb{Q})$ represents the set of all joint distributions $\gamma(\boldsymbol{x}, \boldsymbol{y})$ whose marginals are $\mathbb{P}$ and $\mathbb{Q}$, respectively.

Next, we choose $\mathcal{F} = \{f : \|f\|_{\mathcal{H}} \leq 1\}$ in (4), where $\|\cdot\|_{\mathcal{H}}$ represents a norm defined in the reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ with the reproducing kernel k and then the induced $d_{\mathcal{F}}$ known as MMD. Mathematically, for any valid positive definite symmetric kernel k, there exists an infinite dimensional feature map $\phi\ \mathcal{X} \rightarrow \mathcal{H}$ such that $k(\boldsymbol{x}, \boldsymbol{y}) = \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{y}) \rangle_{\mathcal{H}}$. ϕ is usually denoted as $k(\boldsymbol{x}, \cdot)$ in formal and satisfies the reproducing property

$$f(\boldsymbol{x}) = \langle f, \phi(\boldsymbol{x}) \rangle_{\mathcal{H}} \quad \forall f \in \mathcal{H}. \tag{6}$$

Since we choose $\mathcal{F} = \{f : \|f\|_{\mathcal{H}} \leq 1\}$, then the obtained MMD has the following expression:

$$\begin{aligned} \mathrm{MMD}(\mathbb{P}, \mathbb{Q}) &= \left\{ \sup_{\|f\|_{\mathcal{H}} \leq 1} \left\langle \mathop{\mathbb{E}}_{\boldsymbol{x} \sim \mathbb{P}} \phi(\boldsymbol{x}) - \mathop{\mathbb{E}}_{\boldsymbol{y} \sim \mathbb{Q}} \phi(\boldsymbol{y}),\ f \right\rangle \right\} \\ W &= \| \mathop{\mathbb{E}}_{\boldsymbol{x} \sim \mathbb{P}} \phi(\boldsymbol{x}) - \mathop{\mathbb{E}}_{\boldsymbol{y} \sim \mathbb{Q}} \phi(\boldsymbol{y}) \|_{\mathcal{H}}. \end{aligned} \tag{7}$$

When we select $\|\cdot\|_{\mathcal{H}}$ as ℓ_2-norm and k is a fixed Gaussian kernel, then the resulting MMD is to match all orders of statistics between $\mathbb{P}$ and $\mathbb{Q}$ [34]. In this article, we aim to compare the MMD between the inputs and the reconstructions, so as to add more prior information to train the autoencoder such that we can obtain better unmixing performance. In Section III-B, we will highlight the empirical estimation of MMD in detail, give its explicit computation, and then elaborate on how MMD is embedded into the unmixing model.

C. Basic Operations

The rectified linear unit (ReLU) activation function and batch normalization (BN) [35] are most commonly used operations in the neural network, simple but effective. ReLU enables the gradient propagation simple, and BN can speed up the network training. The ReLU is defined as follows:

$$\varphi(x) = \max(0, x) \tag{8}$$

where $x \in \mathbb{R}$. Given a batch size of vectors $\{\boldsymbol{v}_i\}_{i=1}^m$, the BN can be expressed as

$$\mathrm{BN}(\boldsymbol{v}_i) = \gamma\, \hat{\boldsymbol{v}}_i + \beta \tag{9}$$

where

$$\begin{aligned} \hat{\boldsymbol{v}}_i &= \frac{\boldsymbol{v}_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \\ \mu_{\mathcal{B}} &= \frac{1}{m} \sum_{i=1}^{m} \boldsymbol{v}_i \\ \sigma_{\mathcal{B}}^2 &= \frac{1}{m} \sum_{i=1}^{m} (\boldsymbol{v}_i - \mu_{\mathcal{B}})^2. \end{aligned}$$

γ and β are parameters that need to be learned, and ϵ is a small positive constant. m is the batch size.

III. Methodology

In this section, we first describe the autoencoder architecture used in our model in detail and then depict the empirical estimation of MMD and its explicit computation. Finally, the proposed total loss is released and followed by model analysis.

A. Autoencoder

An autoencoder is a feedforward neural network used to reconstruct the input by means of learning an identical mapping [36], [37]. It is composed of two parts: an encoder compresses the input into a latent low-dimensional representation, and a decoder is employed to transform the representation into an approximation of the input. The autoencoder is trained by minimizing the differences between the original input and the consequent reconstruction. We consider a simple three-layer fully connected autoencoder. In order to speed up and stabilize the training, we add an additional layer into the network, i.e., the BN layer. Next, we are to present the autoencoder architecture of our unmixing model in detail.

Formally, given a batch of spectral vectors with size m, i.e., $\{\boldsymbol{x}_p\}_{p=1}^m \subset \boldsymbol{X}$, the encoding process is to embed the input spectral $\boldsymbol{x}_p$ into abundance vector $\boldsymbol{a}_p$ casted as follows:

$$\boldsymbol{q}_p = \varphi(\mathrm{BN}(\boldsymbol{W}\boldsymbol{x}_p) - \boldsymbol{\alpha}) \tag{10}$$

$$\boldsymbol{a}_p = \mathrm{STO}(\boldsymbol{q}_p) \tag{11}$$

where $\boldsymbol{W} \in \mathbb{R}^{R \times B}$ denotes the encoder weight matrix connecting the input layer with the hidden layer and BN represents the BN operation. $\boldsymbol{\alpha} \in \mathbb{R}^{R \times 1}$ is a learnable parameter utilized to modify the perturbation, which might be caused by average operation of the BN. φ is the ReLU function used to satisfy the ANC, and STO stands for the sum-to-one configuration that is met by the ℓ_1-norm or softmax function. That is,

$$\boldsymbol{a}_p = \ell_1 - \mathrm{norm}(\boldsymbol{q}_p) = \frac{\boldsymbol{q}_p}{(\|\boldsymbol{q}_p\|_1 + \epsilon)} \tag{12}$$

$$\boldsymbol{a}_p = \mathrm{softmax}(\boldsymbol{q}_p) = \frac{e^{\boldsymbol{q}_p}}{\|e^{\boldsymbol{q}_p}\|_1} \tag{13}$$

where ϵ is a very small positive constant to avoid zero-division and set into 10^{-16} under our model. According to the linear

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mixing assumption, the decoding process should be a linear transformation

$$\boldsymbol{y}_p = \boldsymbol{E}\boldsymbol{a}_p \tag{14}$$

where $\boldsymbol{y}_p$ indicates the reconstructed spectra vector and $\boldsymbol{E}$ is the decoder weight matrix that denotes the estimated endmember matrix after the training finished. Here, we use Θ to denote all parameters needed to be trained in the unmixing model, i.e., $\Theta = [\boldsymbol{W}, \boldsymbol{E}, \boldsymbol{\alpha}, \gamma, \beta]$. There generally has a pointwise reconstruction error between $\{\boldsymbol{x}_p\}_{p=1}^m$ and $\{\boldsymbol{y}_p\}_{p=1}^m$ expressed as $(1/m)\sum_{p=1}^m \mathcal{L}(\boldsymbol{x}_p, \boldsymbol{y}_p)$, and minimize this reconstruction error for training. Existing learning-based methods typically adopt the loss functions, such as SAD and mse as $\mathcal{L}$, and they are defined as follows:

$$\mathrm{SAD}(\boldsymbol{x}_p, \boldsymbol{y}_p) = \arccos\left(\frac{\boldsymbol{x}_p^T \boldsymbol{y}_p}{\|\boldsymbol{x}_p\|_2 \|\boldsymbol{y}_p\|_2}\right) \tag{15}$$

$$\mathrm{mse}(\boldsymbol{x}_p, \boldsymbol{y}_p) = \|\boldsymbol{x}_p - \boldsymbol{y}_p\|_2^2. \tag{16}$$

So far, we have completed the construction of our autoencoder architecture.

B. Maximum Mean Discrepancy

Suppose that we have obtained two sets of spectra vectors $\mathcal{D}_s = \{\boldsymbol{x}_p\}_{p=1}^m$ and $\mathcal{D}_t = \{\boldsymbol{y}_p\}_{p=1}^m$ from the inputs (*source* domain) and the reconstructions (*target* domain), respectively. We generally would ask that whether $\mathbb{P}_{\mathcal{D}_s}$ equals to $\mathbb{P}_{\mathcal{D}_t}$, and a simple solution is to compare statistics between these two sets; if they are similar, then these two sets of samples are likely to come from the same distribution. MMD was launched with this in mind. Formally, we denote the square population of MMD measure in (7) as MMD^2, and its empirical estimation is to compute the mean squared difference of the statistics of $\{\boldsymbol{x}_p\}_{p=1}^m$ and $\{\boldsymbol{y}_p\}_{p=1}^m$, which is defined as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{MMD}^2} &= \left\| \frac{1}{m}\sum_{i=1}^m \phi(\boldsymbol{x}_i) - \frac{1}{m}\sum_{j=1}^m \phi(\boldsymbol{y}_j) \right\|_2^2 \\ &= \frac{1}{m^2}\sum_{i=1}^m\sum_{i'=1}^m \phi(\boldsymbol{x}_i)^T\phi(\boldsymbol{x}_{i'}) - \frac{2}{m^2}\sum_{i=1}^m\sum_{j=1}^m \phi(\boldsymbol{x}_i)^T\phi(\boldsymbol{y}_j) \\ &\quad + \frac{1}{m^2}\sum_{j=1}^m\sum_{j'=1}^m \phi(\boldsymbol{y}_j)^T\phi(\boldsymbol{y}_{j'}).\end{aligned} \tag{17}$$

Embracing ϕ as the identity function results in matching the sample mean, and selecting the appropriate ϕ allows for matching higher order moments. By means of applying the kernel trick, (17) can be written as the following expression:

$$\begin{aligned}\mathcal{L}_{\mathrm{MMD}^2} = \frac{1}{m^2}\sum_{i=1}^m\sum_{i'=1}^m k(\boldsymbol{x}_i, \boldsymbol{x}_{i'}) - \frac{2}{m^2}\sum_{i=1}^m\sum_{j=1}^m k(\boldsymbol{x}_i, \boldsymbol{y}_j) \\ + \frac{1}{m^2}\sum_{j=1}^m\sum_{j'=1}^m k(\boldsymbol{y}_j, \boldsymbol{y}_{j'}).\end{aligned} \tag{18}$$

In this article, we employ the Gaussian kernel as the reproducing kernel, which is defined as $k(\boldsymbol{x}, \boldsymbol{y}) = \exp(-(1/2\sigma)\|\boldsymbol{x} - \boldsymbol{y}\|_2^2)$, and σ is the bandwidth parameter. The reason for choosing the Gaussian kernel is that its Taylor expansion contains an infinite number of terms covering all orders of statistics. Minimizing $\mathcal{L}_{\mathrm{MMD}^2}$ under this feature expansion is then equivalent to minimizing a distance between all moments of the two distributions. For ease of calculation, (18) can be expressed as the following matrix operation form:

$$\mathcal{L}_{\mathrm{MMD}^2} = \mathrm{trace}\left(\begin{bmatrix} K_{s,s} & K_{s,t} \\ K_{t,s} & K_{t,t} \end{bmatrix} M\right) \tag{19}$$

where $(K_{s',t'})_{i,j} = \exp(-(1/2\sigma)\|z_i - z_j\|_2^2)$, $z_i \in \mathcal{D}_{s'}, z_j \in \mathcal{D}_{t'}$; $s', t' \in \{s, t\}$, and each element of M is given as follows:

$$M_{ij} = \begin{cases} \dfrac{1}{m^2}, & z_i, z_j \in \mathcal{D}_s \\ \dfrac{1}{m^2}, & z_i, z_j \in \mathcal{D}_t \\ \dfrac{-1}{m^2}, & \text{otherwise.} \end{cases} \tag{20}$$

Up until now, we have completed the details of MMD^2, retrieving from its origin to explicit empirical estimation computation. Next, we will describe how the MMD is integrated into the unmixing model.

C. Loss Function

As mentioned before, our contribution is to combine SAD with MMD as the final loss function for training. The idea is simple and intuitive, adding additional prior information for the autoencoder training and, thus, boosting the unmixing performance. Given the inputs $\{\boldsymbol{x}_p\}_{p=1}^m$ and the reconstructions $\{\boldsymbol{y}_p\}_{p=1}^m$, the proposed total loss is that

$$\mathcal{L}_T = \frac{1}{m \cdot \pi}\sum_{p=1}^m \mathrm{SAD}(\boldsymbol{x}_p, \boldsymbol{y}_p) + \lambda \mathcal{L}_{\mathrm{MMD}} \tag{21}$$

where λ is the tradeoff parameter between underlying loss SAD and regularization term MMD, and $\mathcal{L}_{\mathrm{MMD}}$ is the square root of $\mathcal{L}_{\mathrm{MMD}^2}$ represented as $\sqrt{\mathcal{L}_{\mathrm{MMD}^2}}$. $1/\pi$, here, is to normalize the SAD value to [0, 1]. In Section III-D, we will demonstrate why we choose $\mathcal{L}_{\mathrm{MMD}}$ instead of $\mathcal{L}_{\mathrm{MMD}^2}$. Our goal is to minimize the $\mathcal{L}_T$ to train the proposed autoencoder; the design philosophy of the loss is combing individual difference and overall difference for training, and thus, we can obtain an unmixing model with nicer performance.

Bandwidth Parameter: The choice of the bandwidth parameter σ in the kernel, which is an important setting for determining the statistical efficiency of MMD, is an open problem. There exist some heuristic-based methods to obtain the bandwidth, e.g., a line search [38]. As a simpler approximation, in our experiments, a mixture of K kernels spanning multiple ranges is used. That is,

$$k(\boldsymbol{x}, \boldsymbol{y}) = \sum_{q=1}^K k_{\sigma_q}(\boldsymbol{x}, \boldsymbol{y}) \tag{22}$$

where k_{σ_q} represents a Gaussian kernel with bandwidth parameter σ_q. For the sake of convenience and uniformity, we set $K = 10$ and $\sigma_q \in \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$ in all our experiments.

For sake of understanding the whole proposed unmixing process, we will outline it in Algorithm 1.

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Algorithm 1 PMAE for Hyperspctral Unmixing

Require: Hyperspectral dataset $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p, \ldots, \boldsymbol{x}_P]$, batch size m, regularization parameter λ, and learning rate ℓ_a. Initial autoencoder parameter $\Theta_0 = [\boldsymbol{W}_0, \boldsymbol{E}_0, \boldsymbol{\alpha}_0, \gamma_0, \beta_0]$. Gaussian kernel k with multiple bandwidth parameter σ_q. Tolerance parameter tol and $l = 0$.

while $\frac{\|\boldsymbol{E}_{l+1} - \boldsymbol{E}_l\|_F^2}{\|\boldsymbol{E}_l\|_F^2} > tol$ **do**
  for $i = 1 \cdots [\frac{P}{m}]$ **do**
    Sample $\bar{\boldsymbol{X}}_i = [\boldsymbol{x}_{(i-1)*m+1}, \ldots, \boldsymbol{x}_{(i-1)*m+m}]$;
    Compute $\boldsymbol{A}_i = [\boldsymbol{a}_{(i-1)*m+1}, \ldots, \boldsymbol{a}_{(i-1)*m+m}]$ by equation (11);
    Compute $\bar{\boldsymbol{Y}}_i = [\boldsymbol{y}_{(i-1)*m+1}, \ldots, \boldsymbol{y}_{(i-1)*m+m}]$ by equation (14);
    Compute $\mathcal{L}_{\text{MMD}^2}$ between $\bar{\boldsymbol{X}}_i$ and $\bar{\boldsymbol{Y}}_i$ by equation (18);
    $\mathcal{L}_T = \frac{1}{m}\sum_{j=1}^{m} \text{SAD}(\boldsymbol{x}_{(i-1)*m+j}, \boldsymbol{y}_{(i-1)*m+j}) + \lambda\sqrt{\mathcal{L}_{\text{MMD}^2}}$;
    $\Theta_{l+1} = \text{Adam}(\mathcal{L}_T, \Theta_l, \ell_a)$;
    $l = l + 1$;
  end for
end while
Return: $\boldsymbol{E}_{l+1}$ and $\{\boldsymbol{A}_i\}_{i=1}^{[\frac{P}{m}]}$.

D. Model Analysis

In this section, we are to elaborate on some theoretical guarantees and nice properties of the MMD term in our proposed loss function and, furthermore, explain why we choose $\mathcal{L}_{\text{MMD}}$ instead of $\mathcal{L}_{\text{MMD}^2}$.

Theorem 1 [39]: Given a kernel k, if k is a characteristic kernel, $\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0$ iff $\mathbb{P} = \mathbb{Q}$.

Property 1 [40]: Let empirical measures $p_m := (1/m)\sum_{i=1}^{m} \delta_{X_i}$ and $q_m := (1/m)\sum_{i=1}^{m} \delta_{Y_i}$, where $\delta_{(\cdot)}$ is the Dirac delta function, and $(X_1, \ldots, X_m)$ and $(Y_1, \ldots, Y_m)$ are m-samples from the distributions $\mathbb{P}$ and $\mathbb{Q}$. Then, there has a relationship between $\text{MMD}^2(p_m, q_m)$ and $\text{MMD}^2(\mathbb{P}, \mathbb{Q})$ as follows:

$$\mathbb{E}[|\text{MMD}^2(p_m, q_m) - \text{MMD}^2(\mathbb{P}, \mathbb{Q})|] = O\left(\frac{1}{\sqrt{m}}\right). \tag{23}$$

Theorem 1 and Property 1 demonstrate that the larger the batch size m, the greater the impact of MMD on improving unmixing performance, and the ablation studies in Section IV-C3 validate this view.

Property 2 [40]: Let empirical measures $p_m := (1/m)\sum_{i=1}^{m} \delta_{X_i}$ and $q_m := (1/m)\sum_{i=1}^{m} \delta_{Y_i}$, where $\delta_{(\cdot)}$ is the Dirac delta function, and $(X_1, \ldots, X_m)$ and $(Y_1, \ldots, Y_m)$ are m-samples with data dimensionality d from the distributions $\mathbb{P}$ and $\mathbb{Q}$. Then, there has a relationship between $\text{W}(p_m, q_m)$ and $\text{W}(\mathbb{P}, \mathbb{Q})$ as follows:

$$\mathbb{E}[|\text{W}(p_m, q_m) - \text{W}(\mathbb{P}, \mathbb{Q})|] = O\left(\frac{1}{\sqrt[d]{m}}\right). \tag{24}$$

Property 1 and Property 2 show that the Wasserstein distance would suffer from the sample complexity issues when the data dimensionality d is large, while the MMD does not suffer from it. It should be noted that the hyperspectral data are in very high dimensionality, so taking the MMD as the distribution similarity metric is a more effective way.

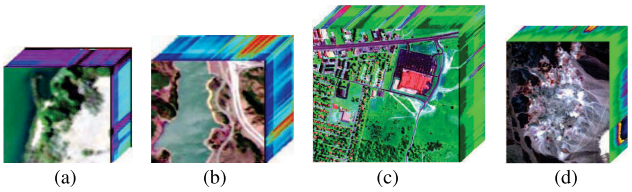

Fig. 1. Cube images of real hyperspectral datasets. (a) Samson. (b) Jasper Ridge. (c) Urban. (d) Cuprite.

Square Root Loss [30]: In the experiments, we prefer to use $\mathcal{L}_{\text{MMD}}$ rather than $\mathcal{L}_{\text{MMD}^2}$ for computing the distribution difference since $\mathcal{L}_{\text{MMD}}$ has advantages in promoting the difference between two distributions to closing to 0 compared with the $\mathcal{L}_{\text{MMD}^2}$. Alternatively, there exists a relationship between $\mathcal{L}_{\text{MMD}}$ and $\mathcal{L}_{\text{MMD}^2}$ as follows:

$$\frac{\partial \mathcal{L}_{\text{MMD}}}{\partial \Theta} = \frac{1}{2\sqrt{\mathcal{L}_{\text{MMD}^2}}} \frac{\partial \mathcal{L}_{\text{MMD}^2}}{\partial \Theta}. \tag{25}$$

The term $1/(2\sqrt{\mathcal{L}_{\text{MMD}^2}})$ is adaptive to the effect of learning rate. This is particularly useful when both $\mathcal{L}_{\text{MMD}^2}$ and $((\partial \mathcal{L}_{\text{MMD}^2})/\partial \Theta)$ are close to 0, where this additional factor can keep larger gradients.

IV. Experiments

In this section, a series of experiments are conducted to demonstrate the effectiveness of our newly proposed PMAE by comparing it against several unmixing methods available in the literature.

A. Data Description

We considered four real hyperspectral datasets to conduct experiments. They are Samon, Jasper Ridge, Urban,[1] and Cuprite[2] that are shown in Fig. 1. Their ground truths can be referred to here.[3] In the following, we provide their detailed descriptions.

1) Samson: This dataset is obtained by the SAMSON sensor with 156 channels, and a subimage with the size of 95×95 is considered. There consists of three constituent materials in the selected scene: Soil (#1), Tree (#2), and Water (#3).

2) Jasper Ridge: This dataset is generated by the AVIRIS sensor, and a subimage of 100×100 is used. To avoid suffering the water vapor absorption and atmospheric influence, the bands 1–3, 108–112, 154–166, and 220–224 are removed and left 198 bands. Four investigated endmembers in this researching scene ate Tree (#1), Water (#2), Soil (#3), and Road (#4).

3) Urban: This dataset has 307×307 pixels, and 162 spectral bands are retained after deleting bands 1–4, 76, 87, 101–111, 136–153, and 198–210. In this observed scene, we examined two cases when the numbers of endmembers

[1] https://rslab.ut.ac.ir/data.
[2] https://aviris.jpl.nasa.gov/data/free_data.html
[3] https://rslab.ut.ac.ir/data.

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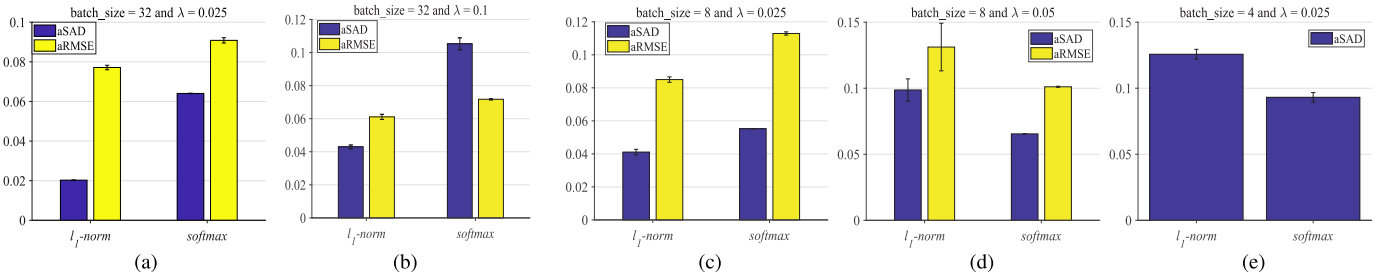

Fig. 2. aSAD and aRMSE results of the five hyperspectral scenes under two sum-to-one configurations. (a) Samson. (b) Jasper Ridge. (c) Urban4. (d) Urban5. (e) Cuprite.

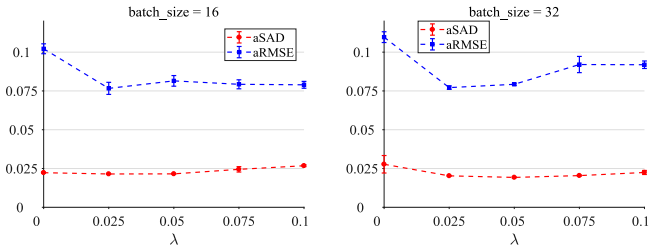

Fig. 3. Ablation study on the Samson dataset.

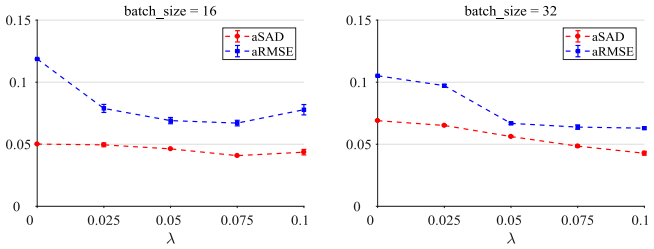

Fig. 4. Ablation study on the Jasper Ridge dataset.

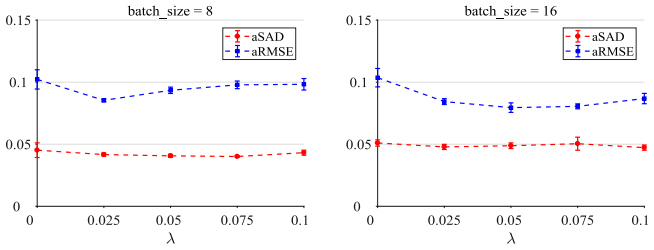

Fig. 5. Ablation study on the Urban4 scene.

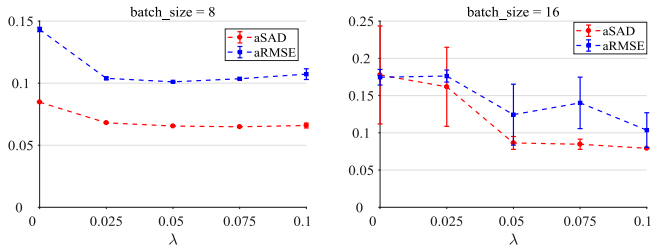

Fig. 6. Ablation study on the Urban5 scene.

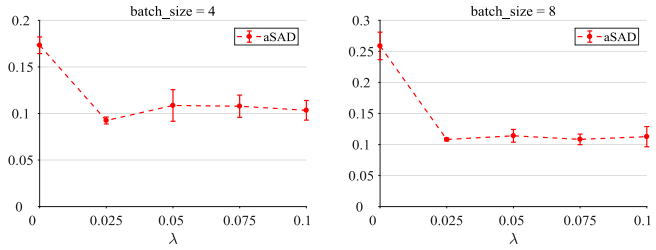

Fig. 7. Ablation study on the Cuprite dataset.

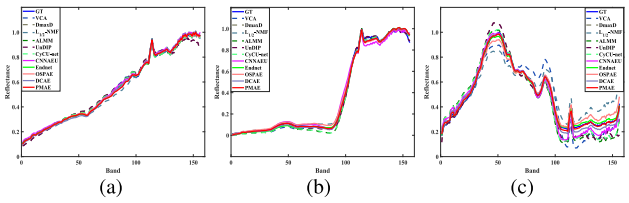

Fig. 8. Estimated endmembers of different methods on the Samson dataset. (a) Soil. (b) Tree. (c) Water.

are 4 and 5. We denoted them as Urban4 and Urban5, that is, *Urban4*: Asphalt (#1), Grass (#2), Tree (#3), and Roof (#4); *Urban5*: Asphalt (#1), Grass (#2), Tree (#3), Roof (#4), and Dirt (#5).

4) Cuprite: We consider investigating its subregion with a size of 250 × 191, and there are 188 channels left after removing the noisy bands (1 and 2 and 221–224) and water absorption bands (104–113 and 148–167); 12 types of unique mineral spectral signatures are held in the scene: Alunite (#1), Andradite (#2), Buddingtonite (#3), Dumortierite (#4), Kaolinite1 (#5), Kaolinite2 (#6), Muscovite (#7), Montmorillonite (#8), Nontronite (#9), Pyrope (#10), Sphene (#11), and Chalcedony (#12).

B. Baselines and Evaluation Metrics

1) Baselines: In order to show that our proposed PMAE method is competitive, ten classical and learning-based unmixing methods are used for comparison.

a) Classical methods: VCA[4] [11], distance-MaxD (DMaxD)[5] [41], $L_{1/2}$ sparsity-constrained NMF ($L_{1/2}$-NMF)[6] [17], and ALMM[7] [20]. Note that VCA and DmaxD are methods for extracting endmembers; we use the method FCLS [12] for abundance estimation after endmember extraction.

[4]http://www.lx.it.pt/bioucas/code.htm
[5]https://sites.google.com/site/robheylenresearch/code
[6]https://github.com/guxizhao/remote.sensing
[7]https://github.com/danfenghong/ALMM_TIP

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TABLE I

QUANTITATIVE RESULTS ($\times 10^{-2}$) OF DIFFERENT UNMIXING METHODS ON THE SAMSON DATASET

End(SAD)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	UnDIP	CyCU-net	CNNAEU	Endnet	OSPAE	DCAE	PMAE
#1	2.52±0.7	4.04±0.0	1.73±0.5	2.15±0.0	4.04±0.0	2.07±0.0	1.89±1.0	1.45±0.1	**0.72±0.0**	1.81±0.8	**1.06±0.0**
#2	4.11±0.2	**2.19±0.0**	3.67±0.2	5.05±0.0	2.19±0.0	4.92±0.0	5.91±0.3	3.35±0.1	4.37±0.4	3.81±0.4	**2.97±0.0**
#3	10.46±38.5	13.04±0.0	11.31±3.3	14.43±0.2	13.04±0.0	12.99±0.0	7.99±1.2	**2.48±0.3**	5.22±1.9	6.82±1.2	**2.05±0.0**
aSAD	15.70±13.2	6.42±0.0	5.57±1.3	7.21±0.0	6.42±0.0	6.66±0.0	5.26±0.6	**2.43±0.2**	3.48±0.8	4.12±0.6	**2.03±0.0**
Abu(RMSE)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	UnDIP	CyCU-net	CNNAEU	Endnet	OSPAE	DCAE	PMAE
#1	25.34±2.1	26.72±0.0	23.51±0.5	12.83±0.0	26.67±0.0	33.91±0.0	**7.39±1.5**	9.49±0.1	13.19±1.2	12.68±1.0	**8.45±0.1**
#2	26.68±2.4	25.15±0.0	19.91±0.5	11.99±0.3	25.21±0.0	18.91±0.0	**5.78±1.7**	7.24±0.1	12.27±1.3	8.91±1.3	**7.23±0.0**
#3	42.91±1.1	42.62±0.0	28.52±0.3	18.73±0.3	42.59±0.0	38.49±0.0	**3.71±0.4**	14.1±0.1	**6.48±0.7**	10.81±1.5	7.39±0.3
aRMSE	31.64±0.7	31.49±0.0	23.98±0.1	14.51±0.2	31.49±0.0	30.44±0.0	**5.63±1.1**	10.27±0.2	10.64±1.1	10.80±0.6	**7.71±0.1**

TABLE II

QUANTITATIVE RESULTS ($\times 10^{-2}$) OF DIFFERENT UNMIXING METHODS ON THE JASPER RIDGE DATASET

End(SAD)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	UnDIP	CyCU-net	CNNAEU	Endnet	OSPAE	DCAE	PMAE
#1	15.17±2.5	15.58±0.0	11.97±2.5	20.71±3.1	14.93±0.0	15.32±0.0	19.31±1.0	6.97±0.4	**3.78±0.8**	12.24±1.8	**2.19±0.1**
#2	32.91±16.2	25.41±0.0	25.58±0.8	23.87±1.9	25.28±0.0	25.54±0.0	15.05±2.7	**3.56±0.9**	4.63±0.3	5.79±3.5	**3.72±0.1**
#3	16.28±4.4	13.36±0.0	17.55±1.5	67.85±9.3	11.44±0.0	11.66±0.0	23.77±5.5	**5.41±0.3**	7.85±3.7	6.24±2.4	**5.36±0.3**
#4	10.15±1.6	10.69±0.0	24.06±9.7	33.51±12.1	27.57±0.0	9.01±0.0	31.57±6.0	**2.61±0.2**	**2.88±0.5**	4.53±1.8	5.91±0.4
aSAD	18.63±5.4	16.23±0.0	19.79±2.7	38.47±5.1	19.81±0.0	15.39±0.0	22.42±2.0	**4.64±0.5**	4.78±1.9	7.20±1.5	**4.29±0.1**
Abu(RMSE)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	UnDIP	CyCU-net	CNNAEU	Endnet	OSPAE	DCAE	PMAE
#1	15.47±0.8	15.99±0.0	18.67±0.2	24.39±5.4	15.91±0.0	18.75±0.0	14.72±0.8	9.11±0.6	**5.86±1.4**	8.13±0.9	**4.95±0.5**
#2	19.11±2.5	20.23±0.0	20.79±3.0	25.93±3.8	14.38±0.0	**3.53±0.0**	11.62±2.0	4.81±0.8	7.91±1.8	7.73±3.8	**4.18±0.1**
#3	13.86±1.3	13.84±0.0	30.83±5.3	37.80±1.9	28.01±0.0	21.93±0.0	16.06±0.6	11.12±0.3	**7.58±0.6**	13.08±0.7	**7.65±0.1**
#4	12.43±1.5	11.18±0.0	27.56±7.3	25.05±1.5	26.35±0.0	18.71±0.0	14.38±0.2	12.94±0.4	**7.15±0.8**	14.68±2.1	**7.73±0.1**
aRMSE	15.22±0.5	15.31±0.0	24.46±2.7	28.95±2.1	21.16±0.0	15.73±0.0	14.20±0.1	9.47±0.6	**7.12±1.1**	10.91±1.1	**6.11±0.2**

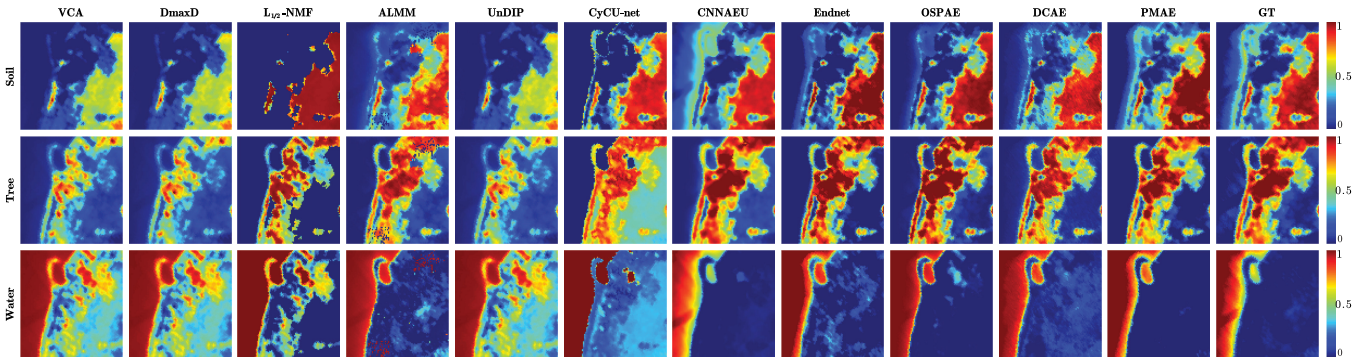

Fig. 9. Visual comparisons of abundance maps among different methods on the Samson dataset.

For the $L_{1/2}$-NMF method, we adopt the VCA-FCLS for the initializations of endmember and abundance matrices.

b) Learning-based methods: Endnet [23], OSPAE [25], DCAE [26], CNNAU [27], UnDIP,[8] [42] and CyCU-net[9] [43]. Their hyperparameter settings are consistent with the original literature. For the OSPAE method, its author provided us their experimental results; thus, we directly referred their reported scores.

2) Evaluation Metrics: In the experiments, four evaluation metrics are utilized to assess the unmixing performance. SAD and root mse (RMSE) are adopted to determine the quality of an individual estimated endmember and abundance, respectively,

$$\mathrm{SAD}(\boldsymbol{e}_k, \hat{\boldsymbol{e}}_k) = \arccos\left(\frac{\boldsymbol{e}_k \cdot \hat{\boldsymbol{e}}_k}{\|\boldsymbol{e}_k\|_2 \|\hat{\boldsymbol{e}}_k\|_2}\right) \tag{26}$$

$$\mathrm{RMSE}(\boldsymbol{m}_k, \hat{\boldsymbol{m}}_k) = \sqrt{\frac{1}{P}\|\boldsymbol{m}_k - \hat{\boldsymbol{m}}_k\|_2^2} \tag{27}$$

where $\hat{\boldsymbol{e}}_k$ and $\hat{\boldsymbol{m}}_k$ are the kth estimated endmember and abundance, respectively, while $\boldsymbol{e}_k$ and $\boldsymbol{m}_k$ are the kth real endmember and abundance. For overall performance or average performance, the average SAD (aSAD) and the average

[8]https://github.com/BehnoodRasti/UnDIP

[9]https://github.com/danfenghong/IEEE_TGRS_CyCU-Net

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TABLE III
QUANTITATIVE RESULTS ($\times 10^{-2}$) OF DIFFERENT UNMIXING METHODS ON THE URBAN4 SCENE

Endmember(SAD)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	CNNAEU	Endnet	DCAE	PMAE
#1	13.75±1.8	13.16±0.0	32.73±8.3	21.23±0.2	14.28±2.4	7.20±0.9	**6.11±0.6**	**6.34±1.0**
#2	>99±31.8	48.18±0.0	24.46±9.1	31.54±5.8	11.87±2.2	6.46±1.9	**3.78±1.3**	**4.43±1.0**
#3	8.19±1.9	7.44±0.0	28.38±20.3	25.85±5.5	8.46±1.2	2.97±0.2	**2.77±0.2**	**2.76±0.4**
#4	21.74±0.0	21.73±0.0	47.34±1.8	76.38±0.4	7.89±3.8	**4.00±0.8**	6.45±1.6	**2.91±0.3**
aSAD	41.17±8.6	22.63±0.0	33.23±4.8	38.73±2.0	10.63±1.1	5.16±0.2	**4.77±0.5**	**4.11±0.2**
Abundance(RMSE)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	CNNAEU	Endnet	DCAE	PMAE
#1	34.09±1.6	39.72±0.0	35.19±1.6	30.21±3.7	14.29±1.1	**11.57±1.0**	16.09±0.9	**9.52±0.2**
#2	41.11±3.6	56.48±0.0	34.92±1.0	35.28±11.2	**14.41±0.3**	15.26±1.6	15.00±1.7	**9.56±0.2**
#3	32.65±2.6	29.22±0.0	25.76±0.9	28.41±1.6	**10.00±0.5**	11.64±0.9	11.03±0.9	**6.98±0.3**
#4	22.93±2.0	21.96±0.0	31.02±0.2	17.35±4.0	**11.81±0.6**	16.55±1.3	14.43±2.1	**7.95±0.8**
aRMSE	32.69±0.7	36.84±0.0	34.23±0.4	27.81±4.1	**12.63±0.2**	13.75±0.7	14.14±0.8	**8.50±0.2**

TABLE IV
QUANTITATIVE RESULTS ($\times 10^{-2}$) OF DIFFERENT UNMIXING METHODS ON THE URBAN5 SCENE

Endmember(SAD)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	CNNAEU	Endnet	OSPAE	DCAE	PMAE
#1	15.27±1.4	14.98±0.0	35.82±15.4	22.50±1.7	24.16±13.9	4.47±0.4	8.11±1.1	**4.36±0.5**	**3.53±0.0**
#2	32.94±26.1	65.69±0.0	15.98±6.2	26.02±5.2	13.89±7.8	5.56±1.0	6.43±1.8	**3.93±1.4**	**5.54±0.1**
#3	11.69±1.7	13.41±0.03	28.17±15.9	27.82±8.2	16.20±1.2	**9.07±0.2**	**6.15±0.5**	10.49±1.6	13.41±0.0
#4	17.40±0.0	7.98±0.0	49.17±2.1	40.62±0.7	13.30±11.7	7.40±0.6	6.01±1.3	**3.80±0.3**	**4.31±0.0**
#5	>99±0.7	12.15±0.0	37.91±26.0	>99±0.7	19.49±1.3	13.34±1.6	**5.17±0.9**	12.79±2.2	**5.93±0.1**
aSAD	39.49±5.6	21.43±0.0	33.41±3.3	46.85±1.8	17.41±0.8	7.97±0.1	**6.83±1.7**	7.07±0.6	**6.55±0.0**
Abundance(RMSE)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	CNNAEU	Endnet	OSPAE	DCAE	PMAE
#1	33.41±1.3	35.35±0.0	32.76±2.3	28.32±3.9	22.00±6.0	**11.69±1.0**	16.55±2.1	17.95±1.0	**11.75±0.4**
#2	40.14±7.9	49.83±0.0	39.00±2.8	34.02±5.3	21.42±1.9	**14.54±0.8**	15.91±1.6	17.84±2.0	**10.72±0.1**
#3	29.62±3.9	29.79±0.0	34.35±4.2	24.38±4.4	13.28±1.5	**11.81±1.2**	15.59±2.3	11.97±1.6	**8.99±0.2**
#4	22.32±0.3	22.57±0.0	27.15±1.6	20.21±3.3	9.53±2.4	15.85±2.0	10.43±1.1	**10.18±1.3**	**7.70±0.3**
#5	39.81±5.8	66.71±0.0	24.95±4.8	25.13±4.0	17.30±2.8	**11.75±1.2**	14.11±1.1	18.95±2.1	**11.39±0.0**
aRMSE	33.06±3.2	40.85±0.0	31.64±2.1	25.94±1.6	16.70±2.3	**13.13±1.0**	14.52±0.6	15.38±0.6	**10.11±0.0**

RMSE (aRMSE) are adopted as follows:

$$\text{aSAD} = \frac{1}{R}\sum_{k=1}^{R}\arccos\left(\frac{\boldsymbol{e}_k \cdot \hat{\boldsymbol{e}}_k}{\|\boldsymbol{e}_k\|_2\|\hat{\boldsymbol{e}}_k\|_2}\right) \tag{28}$$

$$\text{aRMSE} = \frac{1}{R}\sum_{k=1}^{R}\sqrt{\frac{1}{P}\|\boldsymbol{m}_k - \hat{\boldsymbol{m}}_k\|_2^2}. \tag{29}$$

Note that the smaller these four metrics, the better the unmixing performance.

C. Parameter Analysis

In our proposed unmixing model, there are three configurations that impact the final unmixing performance: batch size m, hyperparameter λ, and abundance STO constrain. Next, we will perform ablation experiments on five real hyperspectral scenes to investigate the effects of these three configurations.

1) Experimental Setup: Our approach is implemented on the Tensorflow framework under Python. Our autoencoder unmixing architecture can be referred to Section III-A.

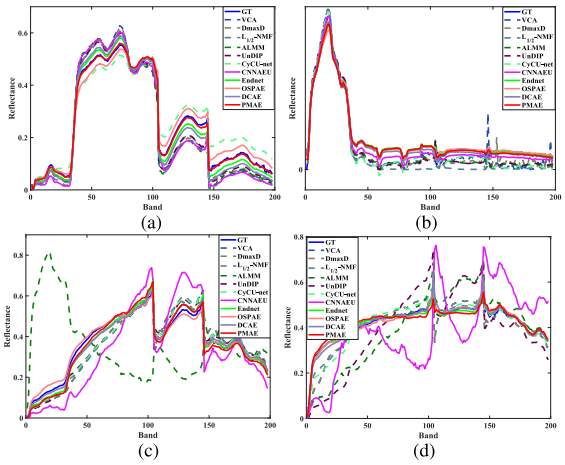

Fig. 10. Endmember comparisons among different methods on the Jasper dataset. (a) Tree. (b) Water. (c) Soil. (d) Road.

The initializations of parameters $[\boldsymbol{W}, \boldsymbol{E}]$ subject to the normal distribution with a standard deviation of 0.01 and initialize $\boldsymbol{\alpha}$ with zeros. γ and β are initialized by zeros and ones.

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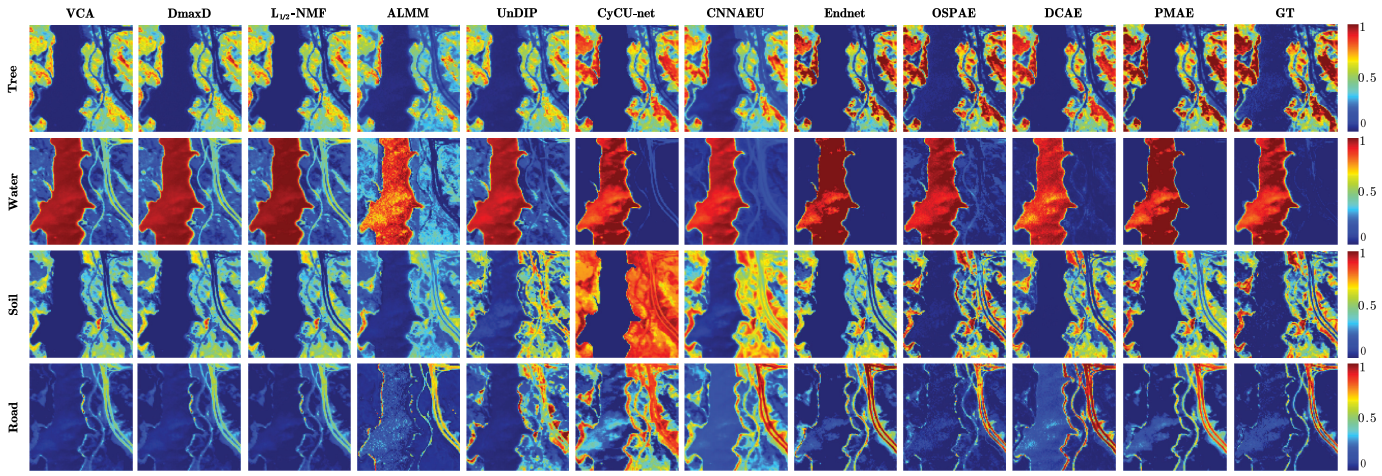

Fig. 11. Visual comparisons of abundance maps among different methods on the Jasper Ridge dataset.

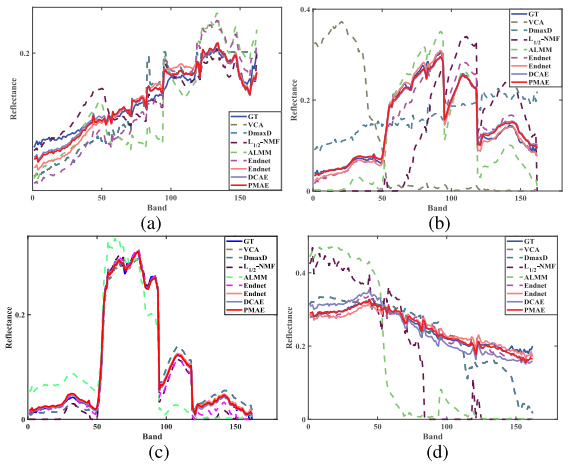

Fig. 12. Endmembers comparisons among different methods on the Urban4 scene. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof.

The Adam optimizer [44] is adopted to update these parameters with default settings, and the learning rate is set to 10^{-4}. The maximum iteration epoch is 100, and our final results are obtained in the epoch when both the aSAD and aRMSE are the smallest. Note that all comparison methods also follow this principle to obtain the final results. All experiments are repeated five times and take the mean value and standard deviation.

2) Abundance Constraints: We first empirically fixed the batch size m and hyperparameter λ and then compare the aSAD and aRMSE results under ℓ_1-norm and softmax configurations, as shown in Fig. 2. From these five bar charts, we can see that, for the Samson, Jasper Ridge, and Urban4, the results obtained by ℓ_1-norm are better than softmax, while, for the Urban5 and Cuprite, the results obtained by softmax are much better than ℓ_1-norm. Therefore, we apply ℓ_1-norm for Samson, Jasper Ridge, and Urban4 while softmax for Urban5 and Cuprite.

3) Ablation Study: Next, we are to investigate the impacts of batch size m and hyperparameter λ on the unmixing performance under these five hyperspectral scenes. The aSAD and aRMSE indicators are used to assess the unmixing performance. We sample m from $\{4, 8, 16, 32\}$ and λ from $\{0, 0.025, 0.05, 0.075, 0.1\}$. Figs. 3–7, respectively, plot the aSAD and aRMSE results for five hyperspectral scenes about batch size m and hyperparameter λ. It can be seen that these five hyperspectral scenes obtain the best results when $(m, \lambda) = (32, 0.025)$, $(m, \lambda) = (32, 0.1)$, $(m, \lambda) = (8, 0.025)$, $(m, \lambda) = (8, 0.05)$, and $(m, \lambda) = (4, 0.025)$, respectively. Note that $\lambda = 0$ means the single SAD loss, while $\lambda \neq 0$ stands for the proposed SAD-MMD loss. From these five trend charts, we can observe that, compared with $\lambda = 0$, the aSAD and aRMSE values at $\lambda \neq 0$ are decreased significantly. When the batch size m is larger, the effect of λ is more obvious. These results demonstrate that the MMD term does play a role in improving unmixing performance. In summary, combining SAD with MMD as the loss function for the autoencoder unmixing model, indeed, can boost the unmixing performance compared with the single SAD loss.

According to the above empirical results, for a real scenario lacking the ground truths, under our unmixing framework, we make a compromise suggestion setting $\lambda = 0.05$. If the number of endmembers is no more than 4, we suggest the STO is ℓ_1-norm, and the batch size is 16. Otherwise, we suggest the STO is softmax and batch size is 8.

D. Performance Evaluation

In this section, we are to quantitatively analyze the experimental results on the five hyperspectral, and these results are evaluated by SAD, RMSE, aSAD, and aRMSE. These quantitative results are summarized in Tables I–V, respectively. The best results are marked in red and the second best in blue.

1) Samson: It can be seen in Table I that the proposed PMAE obtains the lowest aSAD, and CNNAEU gets the smallest aRMSE. The PMAE owns the second smallest aRMSE. As can be seen from Fig. 8, most unmixing methods can easily extract the correct spectra of endmembers since this dataset holds pure pixels. The PMAE enjoys the best result on endmember #3. Fig. 9 provides the visual abundance maps of various unmixing methods. We can see that the abundance maps of CNNAEU and PMAE are closer to the ground truths.

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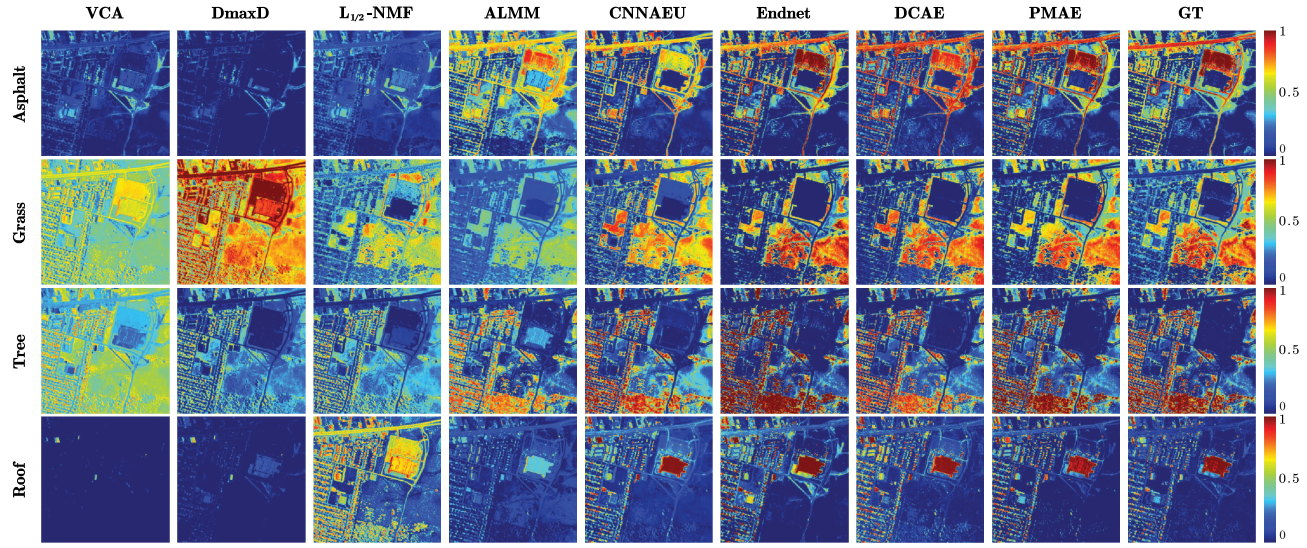

Fig. 13. Visual comparisons of abundance maps among different methods on the Urban4 scene.

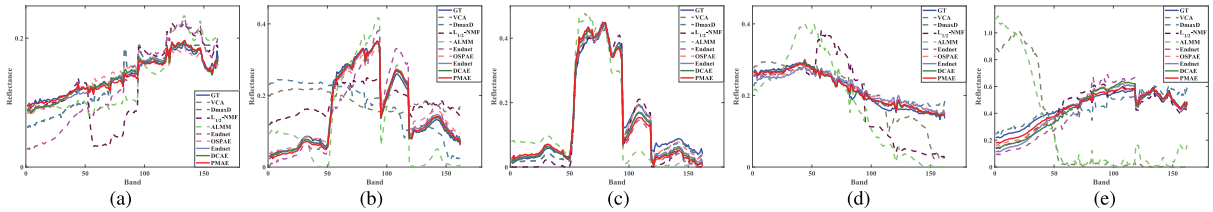

Fig. 14. Endmember comparisons among different methods on the Urban5 scene. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof. (e) Dirt.

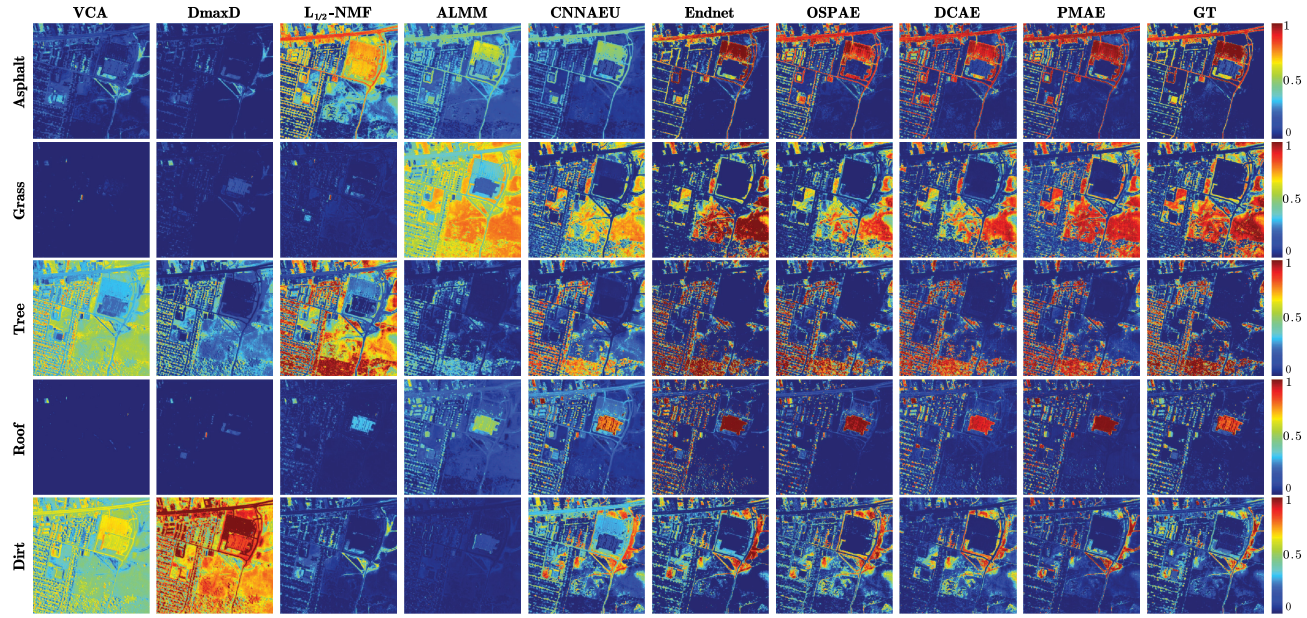

Fig. 15. Visual comparisons of abundance maps among different methods on the Urban5 scene.

2) Jasper Ridge: From Table II, we can see that the proposed PMAE method gets the lowest aSAD and aRMSE. For the individual SAD, the PMAE obtains the best results on endmembers #1 and #3. For the individual RMSE, the PMAE earns the either best or second best results. Figs. 10 and 11 show the endmember and abundance comparisons in visual among different unmixing methods. As shown in Fig. 10, the ALMM method cannot distinguish between endmembers #3 and #4. The endmembers estimated by the remaining methods can match with the ground truths in shape. It can be seen from

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TABLE V

QUANTITATIVE RESULTS ($\times 10^{-2}$) OF DIFFERENT UNMIXING METHODS ON THE CUPRITE DATASET

Endmember(SAD)	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	Endnet	DCAE	PMAE
#1	10.90±2.3	**8.48±0.0**	12.25±1.2	23.10±8.3	**10.00±0.6**	10.44±1.1	11.61±0.7
#2	10.10±2.5	9.96±0.0	9.88±1.3	**6.89±0.7**	8.16±1.0	7.97±0.3	**7.22±0.0**
#3	9.75± 2.1	**7.11±0.0**	**9.23±3.4**	9.43±2.1	24.30±1.5	17.67±5.0	15.07±1.4
#4	9.51±1.9	11.85±0.0	12.91±1.4	13.39±2.6	10.27±0.2	**8.78±0.6**	**7.41±0.4**
#5	11.91±5.1	**8.35±0.0**	11.02±2.0	7.78±0.4	**6.91±0.1**	8.21±0.9	8.44±1.2
#6	8.61±1.6	**8.28±0.0**	11.51±3.6	9.08±1.7	13.25±0.4	6.98±1.5	**6.73±1.1**
#7	12.80±4.0	**10.18±0.0**	11.79±4.4	13.02±3.3	27.90±0.1	15.60±0.8	**10.91±3.6**
#8	7.36±1.4	6.61±0.0	10.31±4.3	7.75±2.0	**6.55±0.0**	7.00±0.3	**6.43±0.3**
#9	7.91±1.2	**7.32±0.0**	8.60±1.3	**7.61±0.2**	9.98±0.1	11.12±3.0	11.14±1.2
#10	8.31±5.2	**5.93±0.0**	9.39±3.8	8.04±2.3	**5.39±0.1**	7.34±1.0	6.66±0.7
#11	21.54±5.1	23.65±0.0	**6.56±1.0**	7.61±5.4	8.30±0.5	6.89±1.3	**6.45±0.9**
#12	**12.34±2.7**	**11.34±0.0**	15.19±2.2	16.00±2.7	14.78±0.3	13.60±0.7	13.71±0.4
aSAD	10.92±1.8	**9.92±0.0**	10.72±1.0	11.01±0.9	12.15±0.3	10.13±0.4	**9.31±0.4**

TABLE VI

QUANTITATIVE RESULTS ($\times 10^{-2}$) OF TWO CONVOLUTIONAL AUTOENCODERS WITH/WITHOUT MMD LOSS IN THE JASPER RIDGE DATASET

End(SAD)	DCAE	DCAE+MMD	CNNAEU	CNNAEU+MMD
#1	12.24±1.8	4.88±2.8	19.31±1.0	8.22±2.3
#2	5.79±3.5	4.25±1.9	15.05±2.7	14.85±4.5
#3	6.24±2.4	3.98±1.7	23.77±5.5	13.11±0.2
#4	4.53±1.8	3.00±0.5	31.57±6.0	20.97±0.5
aSAD	7.2±1.5	4.03±0.6	22.42±2.0	14.29±0.4
Abu(RMSE)	DCAE	DCAE+MMD	CNNAEU	CNNAEU+MMD
#1	8.13±0.9	7.83±0.7	14.72±0.8	6.47±0.6
#2	7.73±3.8	6.87±0.3	11.62±2.0	4.12±0.4
#3	13.08±0.7	11.45±0.8	16.06±0.6	13.14±1.4
#4	14.68±2.1	11.30±1.2	14.38±0.2	14.19±1.3
aRMSE	10.91±1.1	9.36±0.7	14.20±0.1	9.48±0.6

Fig. 11 that the abundance maps of PMAE on materials #1 and #2 are more recognizable, while those of OSPAE on materials #3 and #4 are more visible.

3) Urban4: It can be observed in Table III that the PMAE obtains the smallest aSAD and aRMSE, and the aRMSE is decreased by 32.7% compared to the second best method CNNAEU. For the individual SAD, the PMAE owns the either best or second-best results. For the individual RMSE, the PMAE enjoys the best results, and its every RMSE is correspondingly reduced by 17.72%, 33.66%, 30.2%, and 33.53% compared to the second-best methods. Fig. 12 plots the endmember signatures of different methods, the classical methods perform poorly on predicting the endmember #2 and #4, and the learning-based approaches still maintain a good matching with the ground truths. Fig. 13 shows the visual comparisons of abundance maps among the competing methods. We can see that the classical methods VCA, DmaxD, and $L_{1/2}$-NMF hold indistinguishable results. In terms of profile, all the learning-based approaches are matching with the ground truths. On the level of local details, the proposed PMAE method can predict more accurate abundance maps.

4) Urban5: As can be seen in Table IV, the PMAE again enjoys the smallest aSAD and aRMSE, and the aRMSE is reduced by 23% relative to the second best method Endnet.

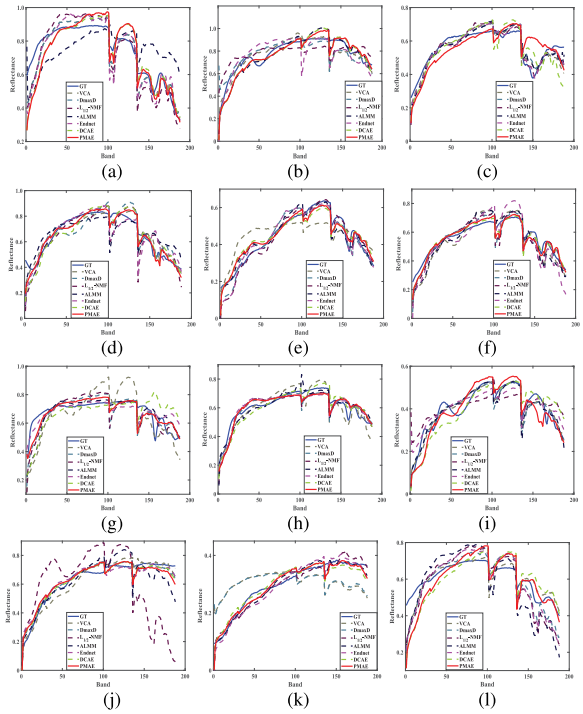

Fig. 16. Endmember comparisons among different methods on the Cuprite dataset. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite$_1$. (f) Kaolinite$_2$. (g) Muscovite. (h) Montmorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony.

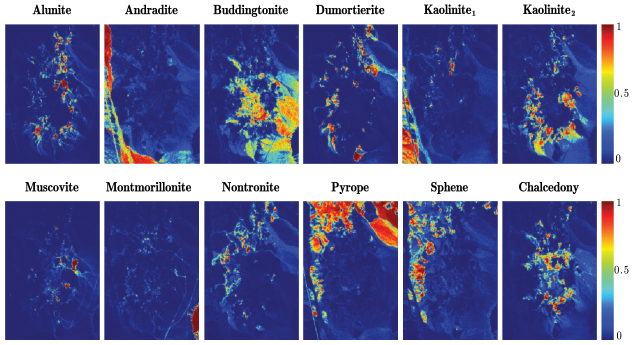

Fig. 17. Visual abundance maps of the Cuprite dataset under the PMAE method.

For the individual SAD, the PMAE is slightly inferior to learning-based methods: OSPAE and DCAE. For the individual RMSE, the PMAE decreases by 25.7% on the material #2, 23.9% on the material #3, and 24.4% on the material #4 compared with the second best methods. The endmember signatures of different unmixing methods are plotted in Fig. 14, and the four classical methods have trouble in estimating the correct endmember #2, but learning-based methods can fit all the ground truths very well. Fig. 15 shows the abundance comparisons among methods, as can be seen that the classical methods estimate the wrong abundance maps. Among the learning-based methods, the CNNAEU gains the worst performance, and the PMAE earns the advantages in predicting abundance maps of materials #2, #3, and #4.

5) Cuprite: Table V shows only the SAD and aSAD performance, and we can observe that all methods are very competitive, but our PMAE method still obtains the lowest aSAD.

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TABLE VII
COMPUTATIONAL TIME (IN SECONDS) OF THE UNMIXING METHODS APPLIED TO THE SAMSON AND JASPER RIDGE DATASETS

Method / Dataset	VCA	DmaxD	$L_{1/2}$-NMF	ALMM	UnDIP	CyCU-net	CNNAEU	Endnet	DCAE	PMAE
Samson	0.35	0.41	33.41	127.17	279.14	104.02	13.06	26.78	76.28	19.75
Jasper Ridge	0.57	0.75	46.71	275.03	359.05	108.08	12.37	29.32	113.63	21.81

Considering the situations both the best SAD and the second best SAD, the DmaxD method deserves to be the best one, but our PMAE is just slightly inferior with it. Fig. 16 presents the endmember comparisons of different methods; it can be seen that all methods can obtain results that fit the ground truths in shape. Fig. 17 gives the visual abundance maps of the PMAE method, as can be seen that there is no overlap between different abundance maps, and every map has high sparsity. These observations demonstrate that our PMAE has the ability to obtain satisfactory unmixing results.

E. Extend to Convolutional Autoencoder

In this section, we insert the proposed SAD-MMD loss to the convolutional autoencoder for clarifying its extensibility on the network architecture. For a fair comparison, we adopt the existing convolutional unmixing autoencoders as the baselines, i.e., DCAE [26] and CNNAEU [27]. In the original papers, they both take the SAD as the loss function. Under these two convolutional autoencoders, we make an unmixing performance comparison on the Jasper Ridge dataset between the SAD loss and the proposed SAD-MMD loss. For the DCAE, we set the batch size to 16 and $\lambda = 0.05$. For the CNNAEU, we set the batch size to 5 and the patch size to 5×5 and $\lambda = 0.05$. Quantitative results of these two convolutional autoencoders with/without MMD loss on the Jasper Ridge dataset are shown in Table VI. As can be seen from this table, under the convolutional autoencoder settings, adding the MMD loss still can improve the unmixing performance a lot when compared with the single SAD loss.

F. Computational Time

In this section, we evaluate the computational cost of different unmixing methods applied to the Samson and Jasper Ridge datasets, and the result of the running time is shown in Table VII. The reported computational times are acquired by using a computer with a 2.9-GHz Intel Core i7-10700 CPU, 16 GB of memory, a 64 bit Operating System, and an NVIDIA GEFORCE RTX 1050Ti GPU with 4-GB memory. The classical methods are implemented in MATLAB R2017a; the learning-based methods are implemented in Python 3.6. In methods of the same type, the proposed PMAE enjoys an acceptable computation cost with nice unmixing performance.

V. CONCLUSION

In this article, we propose an autoencoder based on the probability metric for HU. Specifically, integrate the MMD with the underlying SAD objective as the final loss function to train the unmixing autoencoder. The MMD, which can be seen as a regularization, is designed to draw the distribution similarity between the input spectrums and the reconstructed spectrums by matching their statistics. The SAD is to compare the difference in a pointwise manner, while the MMD considers comparing the overall difference. Combining the individual difference with the overall difference, one can obtain more prior information to guide the autoencoder training; thus, it can boost the unmixing performance. In order to validate this viewpoint, we conducted a series of ablation studies, showing that adding the MMD into the autoencoder unmixing model, indeed, boosts the unmixing performance a lot compared to the single SAD objective. Compared with four classical and six recent learning-based approaches, quantitative and qualitative results under five hyperspectral scenes demonstrate the effectiveness and competitiveness of our method.

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